FORM 2

THE PATENT ACT 1970
(39 of 1970)
&
The Patents Rules, 2003
COMPLETE SPECIFICATION
(See Section 10 and rule 13)

1. TITLE OF THE INVENTION:

An Intelligent Smart Home IoT Security System for Threat Detection, Classification, and Alert Mechanisms

2. APPLICANT(S)

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3. PREAMBLE OF THE DESCRIPTION

COMPLETE
The following specification particular1y describes the invention and the manner in which it is to be performed.

4. Description

5. CLAIMS

6. DATE AND SIGNATURE

7. ABSTRACT OF THE INVENTION

FIELD OF THE INVENTION:

The present invention relates to IoT security and network intrusion detection, specifically to a hybrid machine learning–based framework for smart home environments. Due to limited computational resources and weak built-in security mechanisms, IoT devices are highly vulnerable to cyber threats. The invention provides an intelligent network-level monitoring system that detects anomalous behavior and classifies known attacks in real time by learning normal device communication patterns and identifying behavioral deviations beyond traditional signature-based methods. The system further enables automated threat mitigation and security event notification to enhance protection within smart home networks.

BACKGROUND OF THE INVENTION:

The rapid advancement of Internet of Things (IoT) technology has significantly transformed modern living environments, leading to the widespread adoption of smart home systems incorporating interconnected devices such as smart bulbs, thermostats, ESP32-based controllers, smart plugs, and voice-controlled assistants. These devices continuously communicate over wireless networks to enable automation, remote control, energy efficiency, and enhanced user convenience.

Typically operating over Wi-Fi networks and lightweight communication protocols, IoT devices exchange control commands, sensor readings, authentication signals, and status updates, creating a highly dynamic and continuously active smart home network ecosystem.

In practical deployments, IoT networks are exposed to various cybersecurity threats, including replay attacks involving retransmission of captured packets, brute-force attacks attempting repeated credential combinations, man-in-the-middle (MITM) attacks that intercept and alter communication, denial-of-service (DoS) attacks that disrupt device or network functionality where infected devices participate in coordinated malicious activities. Such threats may compromise device functionality, leak sensitive information, disrupt automation processes, or allow attackers to gain unauthorized control over the smart home network.

Existing IoT security solutions attempt to mitigate these vulnerabilities; however, they exhibit significant limitations. Many systems rely primarily on signature-based intrusion detection techniques that identify attacks using predefined patterns, making them effective only against known threats while remaining incapable of detecting anomalous or previously unseen behavioral patterns.

Additionally, several research approaches depend on simulated datasets that do not accurately reflect real-world traffic behavior, often failing to capture dynamic network fluctuations, device-level constraints, and realistic attack scenarios.

Accordingly, there exists a need for a practical, scalable, and hybrid machine learning–based IoT security framework that integrates anomaly detection, supervised attack classification, performance evaluation, and visualization within a unified architecture. The present invention addresses this requirement by introducing a structured two-stage security model capable of detecting anomalous behavior and classifying known attacks, evaluating multiple machine learning techniques, and presenting actionable results through an interactive monitoring interface tailored specifically for smart home IoT environments.

NOVELTY:

Novel Aspect	**Description**	**Why It's Innovative**
Dataset Generation and Integration	Utilizes both publicly available IoT datasets for baseline study and real-time network traffic captured from the deployed system under normal and attack conditions.	Provides a hybrid dataset that reflects both controlled and practical IoT communication behavior, resulting in more realistic model training, performance validation, and deployment readiness.

Hybrid Two-Stage Detection Framework	Implements a structured two-stage security model that combines unsupervised anomaly detection (LOF, Isolation Forest, One-Class SVM, Elliptic Envelope, Autoencoder) with supervised Random Forest classification and automated alert generation upon threat detection.	Unlike conventional systems that rely on a single detection mechanism, this layered architecture enhances detection accuracy and enables differentiation between normal behavior, known attack behavior, and anomalous network activity.
Behavior-Based Anomaly Detection Framework	Trains anomaly detection algorithms exclusively on normal IoT network traffic to learn baseline communication patterns and identify deviations as anomalous behavior without relying on predefined attack signatures.	Unlike traditional signature-based systems, this approach detects anomalous network behavior based on learned communication patterns, improving adaptability in dynamic smart home environments

OBJECTIVES:

- To provide an intelligent smart home IoT security system configured to continuously monitor communication traffic and behavioral patterns of IoT devices within a home network for early detection of abnormal or suspicious activities.

- To develop an attack evaluation mechanism that applies machine learning–based anomaly detection and classification techniques to identify known and anomalous behaviour.
- To implement an alert framework capable of isolating compromised IoT devices, blocking malicious traffic, preventing lateral threat propagation, and generating real-time security event notifications to users.

SUMMARY OF THE INVENTION:

The present invention provides a hybrid machine learning–based IoT security framework specifically designed for smart home environments. The system continuously monitors network traffic generated by IoT devices, detects abnormal behavior, classifies attacks, and provides an automated alert mechanism.

The invention operates using a structured hybrid two-stage detection framework that combines unsupervised anomaly detection with supervised attack classification to enhance detection accuracy and coverage.

1. Hybrid Two-Stage Detection Framework

Stage 1 – Anomaly Detection (Unsupervised Learning)

In the first stage, the system identifies abnormal network behavior without relying on labeled attack datasets. Machine learning models are trained exclusively on normal IoT network traffic to learn baseline communication patterns. Any deviation from the learned normal behavior is treated as anomalous network behavior, enabling detection of suspicious or irregular activity.

This stage incorporates multiple anomaly detection algorithms, including Local Outlier Factor (LOF) for identifying local density deviations, Isolation Forest for isolating abnormal data points through random partitioning, One-Class Support Vector Machine (SVM) for learning the boundary of normal traffic behavior, Elliptic Envelope for statistical distribution□based anomaly detection, and Autoencoder models that reconstruct normal traffic patterns and flag high reconstruction errors as anomalies.

The multi-model approach enables comparative performance evaluation and selection of the most effective anomaly detection technique tailored to smart home IoT traffic characteristics.

Stage 2 – Attack Classification

Upon detection of anomalous behavior, the system proceeds to classify the specific attack type. In this stage, a supervised machine learning classifier is trained on labeled attack datasets to accurately identify known attack behaviors and determine the nature and category of the detected malicious activity.

The classification stage provides deeper analytical insight by distinguishing between normal behavior, known attack behavior, and anomalous network behavior, thereby supporting informed mitigation decisions and enhancing overall smart home network security.

The system includes an automated alert mechanism that generates real-time notifications when a threat is detected. Alerts are delivered through communication channels such as mobile notifications, electronic mail, or a monitoring dashboard, providing details about the detected attack and affected device to enable timely user response.

2. Performance Evaluation and Model Comparison

The invention incorporates a structured evaluation framework to measure and compare the performance of different machine learning models. Statistical metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis are used to assess detection effectiveness. Accuracy measures overall prediction correctness, precision evaluates the proportion of correctly identified attacks, recall determines the ability to detect actual attack instances, and F1-score provides a balanced measure of precision and recall. The confusion matrix offers a detailed representation of correct and incorrect classifications. This comprehensive evaluation process ensures selection of the most effective model for deployment.

3. Visualization

The system includes a visualization module that displays detection results, model comparison charts, Receiver Operating Characteristic (ROC) curves, and key statistical metrics. The dashboard clearly differentiates between normal and malicious traffic patterns and generates real-time alert notifications upon detection of attack traffic.

BRIEF DESCRIPTION OF THE DRAWINGS

Figure-1: Experimental Smart Home IoT Network Setup

Figure 1 illustrates the experimental smart home IoT network setup used for data collection and traffic monitoring. The setup includes ESP32-based IoT devices, connected smart components, and a monitoring system for capturing real-time network traffic. The figure demonstrates the practical deployment environment used for generating both normal and attack traffic. It highlights the integration of hardware components and traffic capture tools for experimental analysis.

Figure-2: Hybrid Machine Learning–Based IoT Security System Architecture

Figure 2 presents the overall three-tier system architecture of the proposed hybrid machine learning–based IoT security framework. The architecture consists of the Data Tier, Application Tier, and Presentation Tier. It shows the flow of data from traffic capture and preprocessing to anomaly detection, classification, performance evaluation, and alert generation. The figure provides a structured overview of the system modules and their interactions.

Figure-3: Workflow of the Hybrid IoT Security Framework

Figure 3 illustrates the operational workflow of the proposed system. It outlines the step-by-step process including traffic generation, data capture, preprocessing, anomaly detection, attack classification, alert generation, and performance evaluation. The flowchart represents the decision-making logic used to differentiate between normal and anomalous traffic. It also demonstrates how detected anomalies are classified and processed further.

Figure-4: Comparative Performance Analysis of Anomaly Detection Models

Figure 4 shows the comparative performance results of different anomaly detection models used in the system. The figure presents evaluation metrics such as accuracy, precision, recall, and F1-score. It enables analysis of the effectiveness of each model in detecting anomalous behavior. The comparison supports the selection of the most suitable model for deployment.

Figure-5: F1-Score Comparison of Detection Models

Figure 5 illustrates the graphical comparison of model performance using F1-score metrics. The bar chart visually represents the relative performance of each anomaly detection algorithm. This visualization aids in identifying the most accurate and reliable detection technique. It provides a clear comparison for performance-based model selection.

Figure-6: Performance Metrics of the Selected Anomaly Detection Model

Figure 6 presents the detailed performance evaluation of the selected anomaly detection model. It includes metrics such as accuracy, precision, recall, F1-score, and confusion matrix results. The figure demonstrates the model's effectiveness in distinguishing between normal and anomalous traffic. It validates the reliability of the proposed detection framework.

Figure-7: Detection and Classification Output

Figure 7 shows the final detection and classification output generated by the system. It displays the predicted traffic behavior, classified attack type, and associated confidence score. The figure demonstrates how the system differentiates between normal behavior and malicious activity. It reflects the practical output of the hybrid detection and classification framework.

DETAILED DESCRIPTION OF THE INVENTION

The present invention provides a hybrid machine learning–based smart home IoT security framework designed to enable real-time monitoring, anomaly detection, attack classification, and performance evaluation of IoT network traffic within a smart home environment.

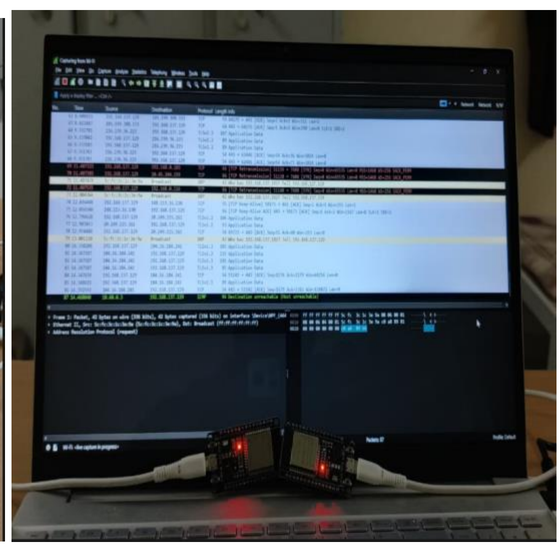
The system operates through a structured three-tier architecture integrating data collection, preprocessing, anomaly detection, supervised attack classification, and evaluation mechanisms into a unified security framework.

Unlike traditional IoT security solutions that primarily rely on encryption or basic protective mechanisms, the proposed invention introduces an advanced behavioral analysis approach capable of identifying abnormal communication patterns and classifying threats using intelligent machine learning algorithms.

The system captures network traffic generated by IoT devices, including ESP32-based smart devices such as smart bulbs, thermostats, and other connected components within the smart home network. The captured traffic data is processed through feature extraction and preprocessing modules to prepare it for analytical evaluation.

Subsequently, the processed data is analyzed to detect deviations from learned normal communication behavior. Any identified anomaly is treated as a potential security threat and is further subjected to classification mechanisms to determine the specific attack type.

This integrated architecture ensures accurate threat detection, intelligent classification, and structured performance evaluation within a practical smart home IoT deployment environment.



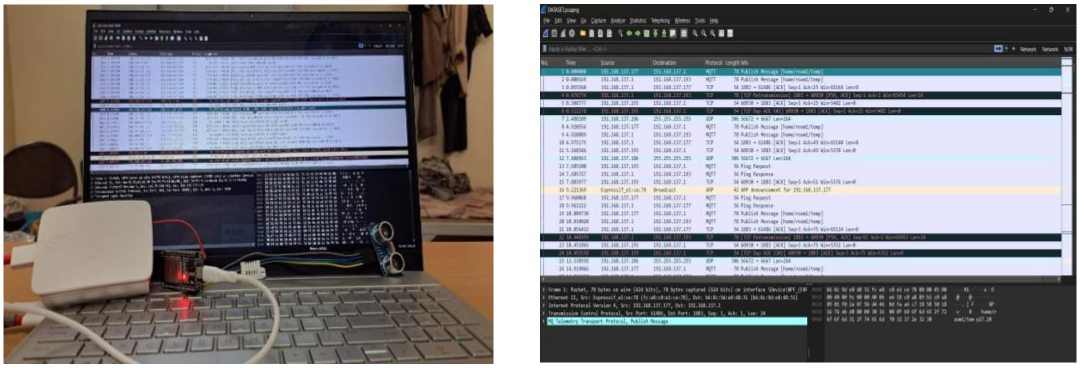


Figure 1: Experimental IoT Network Setup for Traffic Monitoring and Attack Analysis

1. Overview of System Architecture:

The proposed system follows a three-tier architecture consisting of:

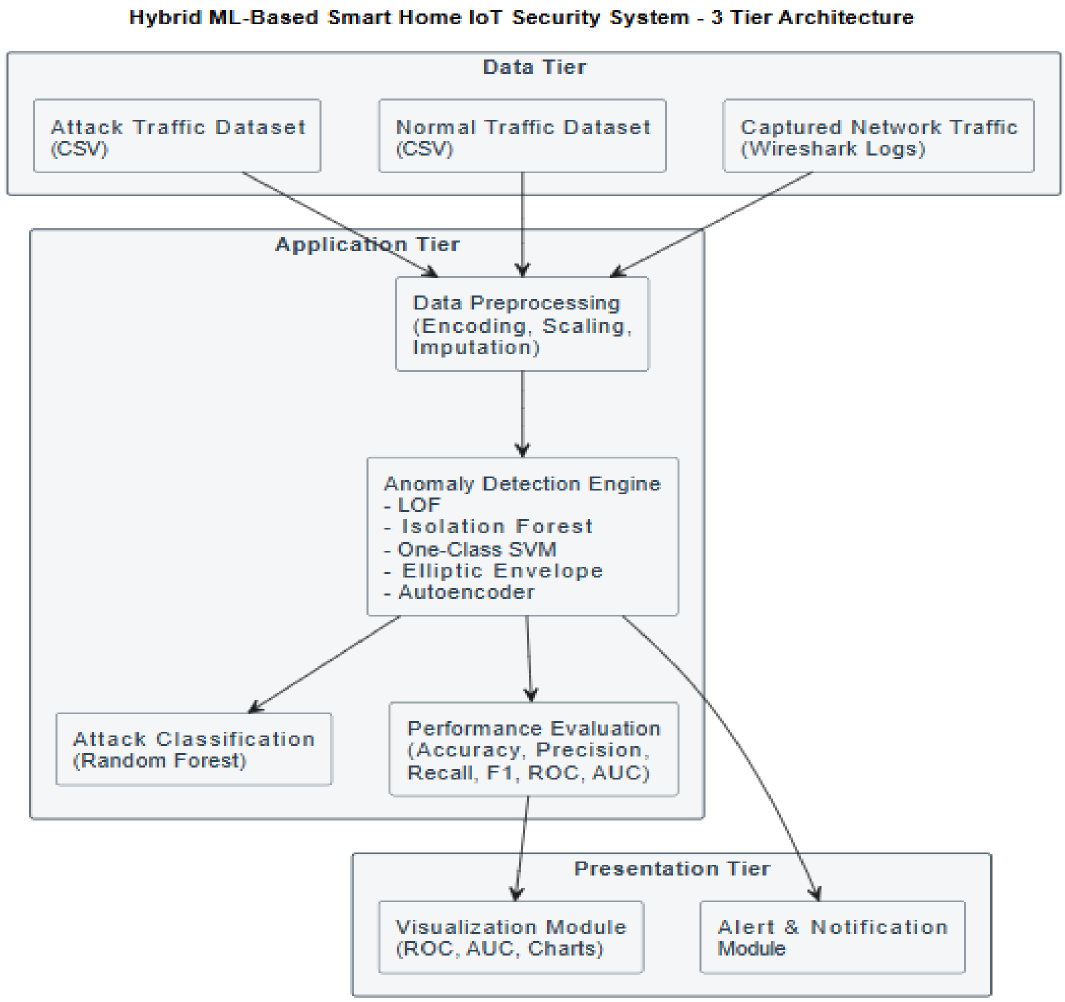


Figure-2: System architecture overview

The system architecture comprises three main layers:

- **Data Layer:**

  The Data Layer is responsible for capturing real IoT network traffic generated within the smart home environment. It stores both normal and attack datasets and maintains structured logs to support analysis, model training, and performance evaluation.

- **Application Layer:**

  The Application Layer performs data preprocessing operations including encoding of categorical features, feature scaling, and handling of missing values. It executes anomaly detection using multiple machine learning models, classifies detected anomalies into specific attack types, and computes performance metrics and statistical evaluation measures to assess model effectiveness.

- **Presentation Layer:**

  The **Presentation Layer** provides an interactive interface for displaying detection results and visual performance comparisons. It generates charts, evaluation summaries, and triggers alert notifications whenever malicious or attack traffic is detected, enabling real-time monitoring and response..

2. System Components:

The invention consists of:

- IoT Devices:
  - Smart home devices such as ESP32 modules generating network traffic.
- Network Monitoring Module:
  - Tools such as Wireshark used to capture real-time packet data.
- Data Preprocessing Engine:
  - Handle:
    - Encoding of categorical features
    - Scaling of numerical values
    - Missing value imputation
- Anomaly Detection Engine:
  - Implements:
    - Local Outlier Factor (LOF)
    - Isolation Forest
    - One-Class SVM

    - Elliptic Envelope
    - Autoencoder
- Evaluation Engine:
  - Calculates:
    - Accuracy
    - Precision
    - F1-score

3. Workflow of the Proposed System:

1. IoT devices generate network traffic.
2. Traffic is captured and stored as structured datasets.
3. Data preprocessing is performed (cleaning, encoding, scaling).
4. Anomaly detection models are trained using normal traffic.
5. Mixed traffic (normal + attack) is evaluated.
6. Abnormal traffic is detected.
7. Detected anomalies are classified into specific attack types.
8. Performance metrics are calculated.
9. Results are visualized and alerts are generated.

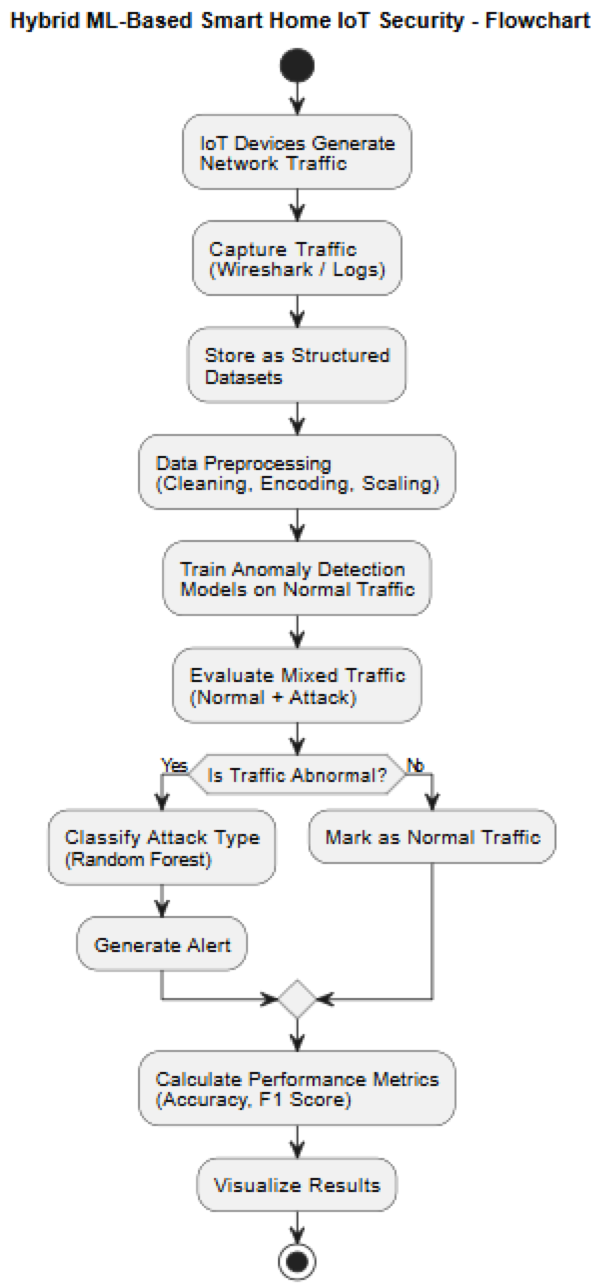


Figure-3: WorkFlow overview

4. Data Collection and Evaluation:

The system collects both normal IoT communication traffic and attack traffic generated during controlled attack scenarios within the smart home environment. These datasets are used to train and evaluate models for distinguishing between normal behavior, known attack behavior, and anomalous network behavior.

The collected data supports training of anomaly detection models, testing of the hybrid detection framework, and evaluation of attack classification accuracy. Performance evaluation includes confusion matrix analysis, Receiver Operating Characteristic (ROC) curve comparison, Area Under the Curve (AUC) measurement, and F1-score comparison across models to determine the most effective detection approach.

Dated this the 10th day of March 2026

CLAIMS
We claim

1. A smart home IoT security system comprising:

   - a network monitoring module configured to continuously monitor communication traffic between one or more IoT devices and a home network;
   - a data preprocessing module configured to extract, clean, and standardize network traffic and device behavioral features;
   - an attack evaluation engine configured to analyze the processed data using machine learning–based anomaly detection and classification techniques to detect malicious activities; and
   - an alert module configured to generate security event notifications to users upon detection of a cyber threat.

2. The system of claim 1, wherein the attack evaluation engine is configured to distinguish between normal behavior, known attack behavior, and anomalous network behavior based on deviations from learned communication patterns of IoT devices.
3. The system of claim 1, wherein the alert module is configured to provide real-time security event notifications through one or more communication channels including mobile notifications, electronic mail, or user interface dashboards.

Dated this the 10th day of March 2026

ABSTRACT

An Intelligent Smart Home IoT Security System for Threat Detection, Classification, and Alert Mechanisms

This invention presents a hybrid machine learning–based IoT security framework designed to detect anomalous behavior and classify known cyberattacks in smart home environments. The system captures real IoT network traffic and applies unsupervised anomaly detection algorithms trained on normal communication behavior to identify deviations from baseline patterns. Multiple detection models, including Local Outlier Factor, Isolation Forest, One-Class Support Vector Machine, Elliptic Envelope, and Autoencoder, are evaluated using statistical performance metrics such as Accuracy, F1-score, Receiver Operating Characteristic (ROC), and Area Under the Curve (AUC). Detected anomalous traffic is further classified using a Random Forest classifier to determine known attack behavior. The framework follows a structured three-tier architecture and provides real-time monitoring, performance evaluation, and automated alert generation. The invention offers a scalable, cost-effective, and intelligent solution for enhancing security in smart home IoT networks.

Model Comparison:

	Model	Accuracy	Precision	Recall	F1
0	IsolationForest	0.494219	0.469442	0.088428	0.148822
1	OneClassSVM	0.632376	0.604677	0.764788	0.675372
2	LOF	0.881991	0.904902	0.853712	0.878562
3	EllipticEnvelope	0.450325	0.006900	0.000695	0.001262
4	Autoencoder	0.781475	0.924557	0.612991	0.737206

Figure-4: Model Comparision

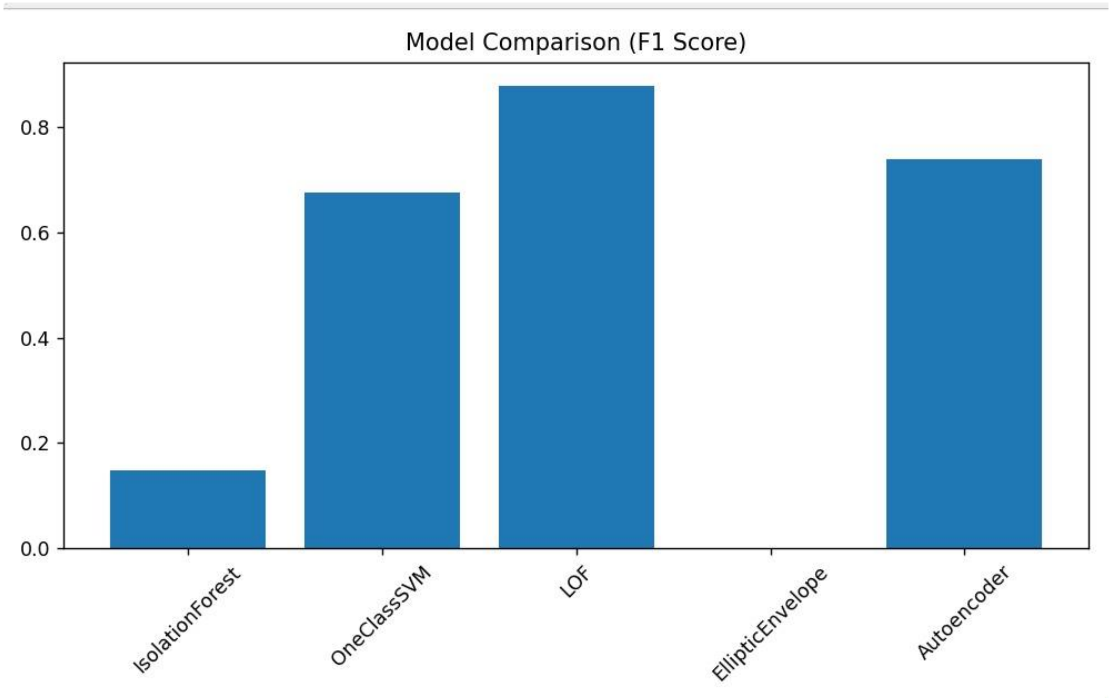


Figure-5: Model F1 Score Comparision

```
=== LOF Detection Performance ===
Accuracy: 0.8819909681901643
Precision: 0.904902167052388
Recall: 0.8537117903930131
F1 Score: 0.8785619446430395
Confusion Matrix:
 [[18342  1808]
 [ 2948 17204]]
```

Figure-6: LOF Model Performance

```
SECURITY ALERT
Prediction: Abnormal Pattern Detected
Email alert sent!
```

Figure-7: Detection and Classification

Dated this the 10th day of March 2026

Applicants: Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology No. of Sheets: 21

Application No.: Sheet No. 1